SURJECTIVITY OF A BASE-CHANGED ALGEBRA MAP IS REALISED AT A FINITE STAGE OF THE COLIMIT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finite generation by finitely many elements. Used at the first line of the counted proof (line 47778). Supplies the finitely many generators x_j of C_0 over B_0 whose images have to be hit.
- Tensor products commute with directed colimits. Used at proof lines 47780-47783. Moves the chosen preimages of the generators down to a single stage of the colimit.

Lemma 0.1. *Let $A = \operatorname{colim}_{i \in I} A_i$ be a directed colimit of rings. Let $0 \in I$ and $\varphi_0 : B_0 \to C_0$ a map of A_0-algebras. Assume*

(1) *$A \otimes_{A_0} B_0 \to A \otimes_{A_0} C_0$ is surjective,*
(2) *C_0 is of finite type over B_0.*

Then for some $i \geq 0$ the map $A_i \otimes_{A_0} B_0 \to A_i \otimes_{A_0} C_0$ is surjective.

Proof. Let $x_1, \ldots, x_m$ be generators for C_0 over B_0. Pick $b_j \in A \otimes_{A_0} B_0$ mapping to $1 \otimes x_j$ in $A \otimes_{A_0} C_0$. For some $i \geq 0$ we can find $b_{j,i} \in A_i \otimes_{A_0} B_0$ mapping to b_j. After increasing i we may assume that $b_{j,i}$ maps to $1 \otimes x_j$ in $A_i \otimes_{A_0} C_0$ for all $j = 1, \ldots, m$. Then this i works. □

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